

Zhanheng Nie

School of Computer Science and Engineering, Beihang University, Beijing, China

NieZH@buaa.edu.cn · +86 19979113056 · [Google Scholar](#)

Education

Beihang University, School of Software

Sep. 2019 – Jun. 2024

B.Eng. in Software Engineering

GPA: 3.82/4.0 (top 5%)

Key Courses: Computer Networks, Data Structures, Algorithm Analysis and Design

Beihang University, School of Computer Science and Engineering

Sep. 2024 – Jun. 2027

M.S. in Computer Science and Engineering Advisor: Prof. Jia Li

GPA: 3.73/4.0 (top 20%)

Key Courses: Matrix Theory, Pattern Recognition, Machine Learning

Publications

* denotes equal contribution.

1. **Zhanheng Nie***, Chenghan Fu*, Daoze Zhang*, Junxian Wu*, et al. MOON2.0: Dynamic Modality-balanced Multimodal Representation Learning for E-commerce Product Understanding. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [Poster]
2. **Zhanheng Nie***, Qisen Wang*, Yifan Zhao, Jia Li. One Step to Realistic: Advancing Infinite Continual Learning with Hierarchical Mixture Architectures. *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, under review.
3. **Zhanheng Nie**, Yifan Zhao, Hongyu Wu, Qigang Wang, et al. HMoLA-Diff: Hierarchical MoE-LoRA for Continual Few-Shot Personalized T2I Generation. *International Conference on Extended Reality (ICXR)*, under review.
4. Daoze Zhang*, Chenghan Fu*, **Zhanheng Nie***, Jianyu Liu*, et al. MOON: Generative MLLM-based Multimodal Representation Learning for E-commerce Product Understanding. *The 19th ACM International Conference on Web Search and Data Mining (WSDM)*, 2026. [Oral]
5. Junxian Wu*, Chenghan Fu*, **Zhanheng Nie***, Daoze Zhang*, et al. MOON3.0: Reasoning-aware Multimodal Representation Learning for E-commerce Product Understanding. *ACM International Conference on Multimedia (ACM MM)*, 2026. [Poster]
6. Chenghan Fu*, Daoze Zhang*, Yukang Lin*, **Zhanheng Nie***, et al. MOON Embedding: Multimodal Representation Learning for E-commerce Search Advertising. Technical Report, *arXiv:2511.11305*, 2025.

Research & Industry Experience

Research Intern, LLM Algorithm Engineer

Apr. 2025 – Present

Alibaba Group, Taobao & Tmall

Search Advertising Algorithm Team

Theme: Scalable multimodal representation learning for industrial-scale product understanding.

- **Full-pipeline multimodal deployment:** Core contributor to deploying MOON across Taobao's search advertising pipeline (recall, relevance, ranking), with a cumulative **+20% CTR** gain, the largest single-project lift in search ads over three years.
- **Generative MLLM for e-commerce representation:** Designed Spatial-Temporal Negative Sampling for MOON, the first generative MLLM-based e-commerce representation model. Pooled same-category hard negatives across distributed GPUs and past batches ($\sim 200\times$ expansion) to improve fine-grained product discrimination. Co-proposed the MBE benchmark.
- **Dynamic modality-balanced representation:** Designed MOON2.0, a multimodal representation framework that replaces fixed-ratio mixed training with joint learning over three modality views (multimodal, image-only, text-only) per sample in a single stage. Proposed Modality-Driven MoE whose Dual-Alignment Matrix binds expert specialization to alignment objectives via sparsity regularization; complemented by Dual-Level Alignment for inter-/intra-product discrimination, MLLM image-text co-augmentation with CLIP filtering, and Dynamic Sample Filtering for noisy pairs. Proposed the MBE2.0 benchmark (released on HuggingFace); achieved SOTA on MBE2.0, M5Product, and Fashion200K in zero-shot evaluation.
- **Reasoning-aware representation with fine-grained enhancement:** Developed the contrastive learning module and FIRE (Fine-grained Residual Enhancement) for MOON3.0, the first reasoning-aware multimodal representation model. FIRE injects gated patch-level residuals and long-range decoder skip connections to

retain fine-grained visual details through the encoding-decoding path. Combined Chain-of-Thought attribute decomposition with joint InfoNCE + GRPO training, where GRPO scores four reward dimensions (format, length, retrieval rank, attribute quality) for structured attribute-level reasoning. Co-proposed the MBE3.0 benchmark.

Graduate Researcher

Sep. 2024 – Present

Beihang University, School of Computer Science and Engineering

Advisor: Prof. Jia Li

Theme: Co-evolution of generation and recognition under resource-constrained continual learning.

- **Infinite continual learning with hierarchical mixtures:** Proposed Online Infinite Continual Learning, a protocol with globally shuffled fine-grained batches and no task boundaries that reveals stability gaps masked by standard CL settings. Designed H-MoMaE, a two-tier Mixture-of-Models + Mixture-of-Experts architecture with hierarchical gating, dynamic sparse routing, and adaptive expansion-pruning. Outperformed 11 baselines across 4 benchmarks; leads by 7+ points on CIFAR-100 under the proposed Joint Accuracy metric.
- **Continual few-shot personalized image generation:** Proposed a hierarchical MoE-LoRA framework for continual few-shot personalized T2I generation. Aligned LoRA expert groups with UNet’s visual hierarchy (object/relation/scene) via cross-modal prototype routing; achieved positive backward transfer through frozen-expert weight-swap distillation with only 4.82M trainable parameters ($\sim 1/180$ of full UNet). The only method among 14 baselines to achieve positive BWT across all task orderings.
- **On-device fine-grained expression recognition:** Collaborated with Huawei on mobile facial expression recognition. Defined a 6-category arousal-based taxonomy, built a T2I + I2I synthetic data pipeline, and developed a TinyViT-based model with structured region attention pooling and multi-subcenter classification; 85.8% accuracy under a 50MB parameter budget.

Open-Source Projects

MBE / MBE2.0 Benchmarks

HuggingFace

Co-designed MBE (2.7M train / 417K test) and MBE2.0 (5.75M train / 636K test), multimodal benchmarks for product representation built from real Taobao product data. Both support zero-shot evaluation across retrieval, classification, and attribute prediction tasks. MBE3.0 is forthcoming.

MOON-1.0 Representation

Open-sourced within Alibaba Group

Core member of the internal release of inference and fine-tuning code, pretrained weights, deployment tutorials, and MBE evaluation datasets; adopted by 10+ teams within Alibaba for product retrieval and search advertising.

Awards & Honors

- Second-Class Academic Scholarship, Beihang University (2025) (top 30%)
- Outstanding Graduate, Beihang University (2024) (top 10%)
- First-Class Freshman Academic Scholarship, Beihang University (2024) (top 30%)
- National Encouragement Scholarship (2022, 2023, 2024) (top 10%)
- University-level Merit Scholarships and “Outstanding Student” honors (6 times, 2019–2024)
- First Prize (Provincial), FLTRP · ETIC Cup English Public Speaking Contest (2022)
- First Prize, National Undergraduate Mathematical Contest in Modeling, Beijing Region (2020, 2022)
- Honorable Mention (H Award), Mathematical Contest in Modeling (MCM), USA (2020)

Skills

- **English:** CET-4 657, CET-6 643 (IELTS planned)
- **Programming Languages:** Python, Shell, C/C++, Java, JavaScript, SQL
- **Frameworks:** PyTorch, Transformers, DeepSpeed, vLLM, FSDP, FlashAttention
- **Model Architectures:** ViT, Diffusion, LoRA, Mixture-of-Experts; InternVL, QwenVL, TBStars-VL
- **Research Interests:** Multimodal Learning, Continual Learning, MLLM Pretraining and Fine-Tuning, Reinforcement Learning, Fine-grained Visual Understanding, Image Generation